Customer Churn Intelligence in Banking

FinalProjectReport | DataVisualization&Analytics

Sector: Banking & Financial Services

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GitHub Repository: Link

Tableau Public Dashboard: Link

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2. Executive Summary

Problem

Customer churn is a major concern in the banking sector as it directly impacts revenue and long-term customer value. Banks lose substantial income when customers close accounts and switch to competitors — and the cost of acquiring a replacement customer far exceeds that of retaining an existing one. This project addresses the challenge of identifying the key behavioural and financial factors that drive churn, enabling early intervention before customers leave.

Approach

The project followed a structured data analytics pipeline. A customer-level banking dataset of approximately 28,000 records was sourced, cleaned, and pre-processed using Python. Derived KPIs including Churn Rate, Active Rate, Average Balance, and Balance Change were computed from raw fields. Exploratory data analysis (EDA) was performed to uncover patterns across activity status, tenure, balance trends, and risk segments. Three interactive Tableau dashboards were built to translate these findings into accessible decision-support tools.

Key Insights

- Inactive customers churn at significantly higher rates than their active counterparts.
- New customers with 0–2 years of tenure represent the highest churn risk cohort.
- Customers with declining account balances show a strong propensity to churn.
- High-risk customer segments exhibit disproportionately elevated churn rates across all metrics.

Key Recommendations

- Deploy targeted early engagement campaigns for inactive customers and new joiners within their first year.
- Implement automated balance-decline alerts to trigger proactive outreach before attrition occurs.
- Develop personalized retention strategies for high-risk segments using the risk segmentation framework built in this project.

4. Problem Statement & Objectives

Formal Problem Definition

What key customer behavioural and financial factors influence churn in a banking system, and how can high-risk customers be identified early enough for effective retention intervention?

Objectives

1. Identify the primary drivers of customer churn through structured exploratory analysis.
2. Segment customers by risk level to prioritize retention resources.
3. Analyze the financial behavior patterns of churned vs. retained customers.
4. Build interactive Tableau dashboards that enable non-technical stakeholders to explore churn trends dynamically.

Project Scope

- In Scope: Descriptive and diagnostic analytics on customer churn using the provided dataset; KPI computation; EDA; Tableau dashboard development.
- Out of Scope: Real-time data pipelines, live CRM integration, and machine learning-based churn prediction models (identified as future scope).

Success Criteria

- Clear, evidence-based identification of the top churn drivers.
- Actionable insights that map directly to retention strategies.
- Three fully functional, publicly accessible Tableau dashboards with interactive filters.

5. Data Description

Dataset Source: [Insert dataset source citation and direct access link]

Number of Records: Approximately 28,000 customer-level rows

Time Period: [Inserttime period covered by dataset]

Key Fields

Field Name	Data Type	Description
Churn Label	Binary	Indicates whether the customer churned (1) or was retained (0)
Activity Status	Categorical	Whether the customer is Active or Inactive based on recent transactions
Tenure (Years)	Numeric	Number of years the customer has held an account with the bank
Account Balance	Numeric	Current or average account balance in currency units
Risk Segment	Categorical	Customer risk classification (e.g., Low, Medium, High)
Age	Numeric	Customer age in years
Occupation	Categorical	Customer's reported occupation category
Gender	Categorical	Customer's gender

Data Limitations & Known Biases

- The dataset does not include time-series tracking of individual customer behavioral changes over time, limiting dynamic trend analysis at the micro level.
- Behavioural features are limited; richer signals such as transaction frequency, product usage depth, or complaint history are not available.
- The churn label is assumed to be accurately assigned; mislabeling in source data would propagate through all analyses.
- Demographic representation may not be uniformly distributed, which could introduce bias in segment-level conclusions.

6. Data Cleaning & ETL Pipeline

All primary data cleaning and transformation steps were executed in Python, in accordance with Capstone 2 requirements. The cleaned dataset was then exported to Tableau for visualization. The full cleaning logic is documented in notebooks/02_cleaning.ipynb, committed to the project GitHub repository.

Missing Values

- Missing values were identified across relevant columns using pandas isnull() diagnostics.
- Columns with small proportions of missing values were imputed using median (numeric) or mode (categorical) strategies.
- Rows with missing churn labels the primary target variable were excluded from analysis to preserve integrity.

Outlier Detection & Treatment

- Numeric fields (Account Balance, Age, Tenure) were assessed using IQR-based outlier detection.
- Extreme outliers were capped at the 1st and 99th percentile boundaries (Winsorization) to limit distortion in aggregate metrics without data loss.

Data Type Conversions & Standardization

- Churn labels were standardized to binary integer format (1 = Churned, 0 = Retained).
- Activity Status values were normalized to a consistent categorical encoding (Active / Inactive).
- Date fields, where present, were converted to datetime format for tenure computation.
- Categorical columns were stripped of leading/trailing whitespace and converted to title case.

Feature Engineering

- Churn Rate: Computed as the proportion of churned customers within each defined segment or cohort.
- Activity Status Flag: Derived binary field used for segmentation analysis.
- Balance Difference: Calculated as the delta between opening and closing balance where both are available, used as a proxy for financial trajectory.
- Tenure Band: Customers were grouped into tenure cohorts (0–2 years, 3–5 years, 6–10 years, 10+ years) for cohort-level analysis.

Assumptions Made During Cleaning

- Customers with zero balance who are active are treated as genuinely active unless the churn label confirms otherwise.
- Imputed values are assumed to represent the central tendency of the respective segment.
- Winsorized outlier boundaries are assumed to represent the operational range of valid customer data.

7. KPI & Metric Framework

The following KPIs were defined to measure churn risk and customer health across the dataset. Each metric is directly tied to the project's business objectives and is computable from the cleaned dataset.

KPI	Formula / Computation	Business Relevance
Churn Rate	(Churned Customers / Total Customers) × 100	Primary metric; measures the share of customers lost within a segment or period.
Active Rate	(Active Customers / Total Customers) × 100	Indicates engagement health; low active rate correlates with elevated churn risk.
Average Balance	Mean of Account Balance across cohort	Proxy for financial engagement; declining balances signal disengagement.
Balance Change	Closing Balance − Opening Balance	Captures financial trajectory; negative values indicate potential pre-churn behavior.
Churn by Segment	Churn Rate computed within each Risk / Tenure / Occupation group	Enables targeted prioritization of retention resources by segment.

All KPI computations are implemented in notebooks/03_eda.ipynb and notebooks/04_statistical_analysis.ipynb. Calculated fields in Tableau reference the exported computed dataset.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to surface patterns in customer behavior and financial characteristics that explain churn. The full code and charts are available in notebooks/03_eda.ipynb.

Activity Status vs. Churn

Analysis of activity status reveals a stark divergence: inactive customers churn at a rate significantly higher than active customers. This confirms that disengagement measured by reduced transactional activity — is one of the strongest leading indicators of churn. Active customers not only exhibit lower churn rates but also maintain higher average balances, reinforcing the value of engagement as both a retention signal and a financial health indicator.

Tenure vs. Churn (Cohort Analysis)

Customers in the 0–2 year tenure band demonstrate the highest vulnerability to churn. This pattern is consistent with the well-documented "early-stage attrition" phenomenon in financial services, where customers who have not yet formed strong product habits or banking relationships are more susceptible to competitive offers or dissatisfaction. Customers with tenure above 5 years show markedly lower churn rates, suggesting that longevity builds stickiness.

Balance Trends vs. Churn

Customers who churned exhibit a negative balance trajectory declining average balances in the period preceding churn — compared to retained customers who maintain stable or growing balances. This makes balance change a highly actionable early-warning signal: a sustained decline in balance can trigger proactive outreach before the customer formally exits.

Risk Segmentation vs. Churn

High-risk customers, as classified in the dataset's risk segmentation field, exhibit disproportionately elevated churn rates relative to low- and medium-risk segments. This finding validates the utility of the existing risk classification as a targeting criterion for retention campaigns and suggests that high-risk customers should be the primary focus of intervention resources.

Demographic Distribution

Age group analysis reveals differential churn rates across life stages, with younger customer cohorts showing higher churn propensity. Occupation-level breakdowns indicate that certain job categories correlate with higher churn rates, potentially reflecting income stability or financial sophistication differences. Gender was found to have a relatively weaker standalone effect on churn compared to behavioural indicators.

9. Statistical Analysis

Statistical techniques were applied to validate patterns observed during EDA and to provide deeper quantitative grounding for the insights. Full implementation is in notebooks/04_statistical_analysis.ipynb.

Segmentation Analysis

Customers were segmented into cohorts based on tenure band, activity status, risk classification, and balance trajectory. Churn rates were computed independently for each cohort, enabling cross-segment comparison. Segments with statistically distinct churn rates were identified as priority targets for intervention.

Hypothesis Testing

A chi-squared test of independence was applied to assess whether the relationship between activity status and churn is statistically significant. Results confirmed a significant association ($p < 0.05$), validating the observed pattern as non-random. Similar hypothesis testing was conducted for risk segment classification versus churn outcome.

Correlation Analysis

Correlation analysis between numeric variables (Balance, Tenure, Age) and the binary churn outcome was performed using point-biserial correlation. Account balance and tenure showed the strongest negative correlations with churn, meaning customers with higher balances and longer tenure are significantly less likely to churn — supporting both KPIs as strong predictors.

Distribution Analysis

Balance and tenure distributions were examined separately for churned and retained customers. Churned customers exhibit a left-skewed balance distribution (concentrated at lower balances) and a right-skewed tenure distribution (concentrated in early years), while retained customers show the opposite. These distributional differences confirm the directional findings from the EDA.

Anomaly Detection

Customers with anomalously low balances combined with inactive status were flagged as high-churn-risk anomalies. These edge-case customers represent an immediate intervention opportunity and were incorporated into the high-risk segment definition used in the Tableau dashboards.

10. Tableau Dashboard Design

Three Tableau dashboards were designed to translate analytical findings into interactive, decision-ready tools. Each dashboard serves a distinct business question and audience.

Tableau Public URL: [Insert Tableau Public Dashboard URL here]

Dashboard 1: Behavior Analysis Dashboard

Purpose: Answer the question "Why are customers churning?" by surfacing behavioral signals.

- Activity vs. Churn: Side-by-side comparison of churn rates for active and inactive customer groups.
- Tenure vs. Churn: Bar chart showing churn rate by tenure cohort (0–2, 3–5, 6–10, 10+ years).
- Balance Decline Analysis: Visualization of average balance trend for churned vs. retained customers over time.
- Filters: Activity Status, Tenure Band, Gender enabling interactive drill-down.

Dashboard 2: Customer Segmentation Dashboard

Purpose: Answer "Who is at risk?" by enabling segment-level churn exploration.

- Risk Segmentation Breakdown: Churn rate distribution across Low, Medium, and High risk segments.
- Age Group Analysis: Churn rate heat map or bar chart by age cohort.
- Occupation Breakdown: Churn proportions across major occupation categories.
- Filters: Risk Segment, Age Group, Occupation supporting targeted campaign planning.

Dashboard 3: Customer Value & Balance Dashboard

Purpose: Answer "What is the financial impact?" by linking churn behavior to balance dynamics.

- Balance Distribution: Histogram comparing balance distributions of churned and retained customers.
- Balance Change Analysis: Visualization of balance deltas for pre-churn vs. retained cohorts.
- Average Balance by Segment: Cross-tabulation of average balance across risk and activity segments.
- Filters: Balance Range, Risk Segment enabling financial impact quantification.

All dashboard screenshots are committed to tableau/screenshots/ and links are documented in tableau/dashboard_links.md in the project GitHub repository.

#	Insight	Business Implication
1	Inactive customers churn at rates significantly above the overall average.	Inactivity is the single most reliable leading indicator of churn; monitoring activity levels enables earlier intervention.
2	Customers in the 0–2 year tenure band have the highest churn probability.	Onboarding quality and early engagement directly determine long-term retention; the first two years are the critical retention window.
3	Declining account balances precede churn in a majority of cases.	Balance trajectory is an early warning signal; automated alerts on sustained balance drops can trigger proactive outreach.
4	High-risk segment customers exhibit disproportionately higher churn rates.	Risk classification is a valid and actionable targeting criterion for concentrated retention investment.
5	Active customers maintain meaningfully higher average balances than inactive ones.	Engagement and financial health are co-indicators; increasing activity rates is likely to improve balance retention.
6	Certain occupation categories show elevated churn rates.	Occupation-specific campaigns (e.g., for freelancers or gig workers with irregular income) may address unique financial instability drivers.
7	Younger age cohorts display higher churn propensity than older customer groups.	Younger customers may respond better to digital engagement tools, flexible products, and incentive-based loyalty programs.
8	Customers with both inactive status and low balance represent a concentrated, immediately actionable churn-risk cluster.	This dual-criteria filter produces a high-confidence at-risk list that should be the first target of any retention campaign.

12. Recommendations

The following five recommendations are prioritized by expected business impact, feasibility, and alignment with the insights derived from this analysis. Each maps directly from insight to recommended action to expected outcome.

Recommendation 1: Deploy Early Engagement Campaigns for New Customers

Insight: New customers (0–2 years) exhibit the highest churn rate.

Action: Design a structured onboarding journey with 30, 60, and 90-day engagement touchpoints including personalized product recommendations, financial health check-ins, and loyalty incentives for early product adoption.

Expected Impact: Reduction in early-stage churn; improved activation rates for new customer cohorts.

Feasibility: Moderate — requires CRM workflow setup and content development but is executable within existing infrastructure.

Recommendation 2: Implement Automated Balance-Decline Alerts

Insight: Declining balances reliably precede customer churn.

Action: Configure rules-based triggers in the banking platform that flag customers with sustained balance declines (e.g., >20% drop over 60 days) and automatically route them to relationship managers for outreach.

Expected Impact: Earlier detection of pre-churn behavior; increased save rate from proactive contact.

Feasibility: High — most core banking systems support rule-based triggers; minimal custom development required.

Recommendation 3: Reactivation Program for Inactive Customers

Insight: Inactive customers churn at rates far exceeding the baseline.

Action: Launch a targeted reactivation campaign for customers flagged as inactive for 60+ days offering incentives (fee waivers, rate improvements, bonus rewards) conditional on completing a qualifying transaction.

Expected Impact: Conversion of a meaningful share of inactive customers to active status; reduction in overall churn rate.

Feasibility: High campaign mechanics are straightforward; ROI is easily measurable.

Recommendation 4: Personalized Retention for High-Risk Segments

Insight: High-risk customers are the most concentrated churn risk pool.

13. Impact Estimation

The following impact estimates are directional approximations based on the dataset analysis and industry benchmarks. Exact figures will depend on implementation scope, campaign execution quality, and market conditions. The logic is presented transparently to allow stakeholders to adjust assumptions.

Recommendation	Estimated Churn Reduction	Business Impact
Early Engagement for New Customers	10–20% of early-stage churn	Stabilizes the customer base; improves LTV of new cohorts.
Automated Balance-Decline Alerts	5–15% of balance-driven churn	Saves revenue from pre-churn customers at low operational cost.
Inactive Customer Reactivation	8–15% reactivation rate	Converts dormant accounts to revenue-generating relationships.
High-Risk Segment Retention	15–25% of high-risk churn	Highest absolute impact given concentration of risk in this segment.
Occupation-Specific Products	5–10% in targeted segments	Opens new revenue streams while reducing financial stress-driven exits.

Industry data suggests that even a 5% reduction in overall churn can increase annual profits by 25–95% (Bain & Company benchmark), given the compounding nature of customer lifetime value. The stakeholder case for acting now is strong: the longer at-risk customers remain unaddressed, the higher the probability of irreversible attrition.

16. Conclusion

Customer churn in banking is a high-stakes, measurable, and addressable problem. This project tackled the challenge of identifying the behavioural and financial factors that drive churn across a dataset of approximately 28,000 customers using a rigorous Python-based analytics pipeline and interactive Tableau dashboards.

The analysis revealed that inactive customers, new customers in the 0–2 year tenure band, and customers with declining balances represent the highest-risk cohorts findings that are both statistically validated and intuitively actionable. High-risk segment classification proved to be a reliable targeting lens for concentrated retention investment.

The five recommendations centered on early engagement, automated balance-decline detection, inactive customer reactivation, personalized high-risk retention, and occupation-specific product development are prioritized by feasibility and expected impact. Even conservative implementation of these strategies is projected to yield meaningful reductions in churn and corresponding improvements in customer lifetime value and profitability.

The three interactive Tableau dashboards provide a durable analytical asset that business teams can use on an ongoing basis to monitor churn risk, explore segment dynamics, and track the impact of retention initiatives over time. This project delivers not just insight, but infrastructure for sustained, data-driven customer retention.

18. Contribution Matrix

This matrix documents each team member's contribution across all project phases. Claims are verifiable through GitHub Insights, PR history, and committed files.

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT & Viva
Lakshya Bapna (2401010252)	✓	✓			✓		
Aarya Srivastava (2401010008)			✓		✓		
Mayank Yadav (2401010271)		✓					✓
Sarvjeet Yadav (2401010424)			✓				✓
Omkar Hadole (2401010175)					✓		✓
Debasish Karan (2401010141)				✓			✓

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted deliverables.